

DPEN021: Enabling Mathematics A

Class Questions Week 6

Week 6A

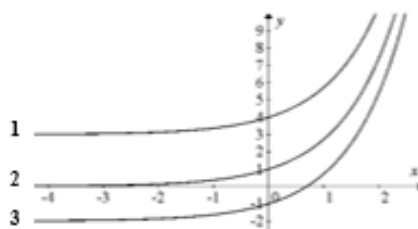
Key Idea - Solving Equations with Exponentials

- Solve the following equations, give your answer in exact form.
 - $e^x = 12$
 - $5^x = 200$
 - $10^x = 90$
 - $4^x - 3 = 12$
 - $10^{5x} = 4$
 - $e^{-4x} + 1 = 9$
 - $2 \times 3^{5-3x} = 1$
 - $2^{3x-1} + 3 = 22$
 - $1 + 3e^{2x} = 28$
 - $2 \times 3^x - 10 = 3^x$
 - $4e^{2x} - 6 = e^{2x}$
 - $4^x + 4^{x+1} = 15$
- Solve the following equations, give your answer correct to 3 decimal places.
 - $10^x = 50$
 - $e^x = 100$
 - $e^{3x} - 3 = 2$
 - $5e^{2x} = 10$
 - $2 \times 10^{5-3x} = 1$
 - $1 + 10^{2x-1} = 7$
 - $4e^{4x+1} + 1 = 9$
 - $3 \times 10^{3x-1} - 7 = 10^{3x-1}$
 - $5e^{2x} - 6 = e^{2x}$

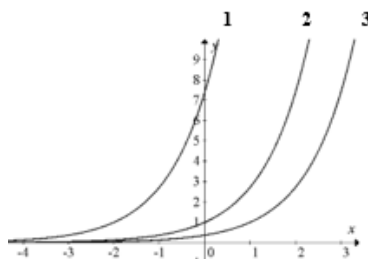
Key Idea - Graphs of Exponential Functions

- For each of the following match the equation of the function with its graph.

- $y = e^x$
 - $y = e^x - 2$
 - $y = e^x + 3$



- $y = e^x$
 - $y = e^{x+2}$
 - $y = e^{x-1}$



- Compared to the graph of $y = e^x$, the graph of $g(x)$ is shifted down 3 units. Write the equation of $g(x)$.
 - Compared to the graph of $y = e^x$, the graph of $g(x)$ is shifted to the right by 3 units. Write the equation of $g(x)$.
 - Compared to the graph of $y = e^x$, the graph of $g(x)$ is shifted 4 units upwards and 5 units to the left. Write the equation of $g(x)$.

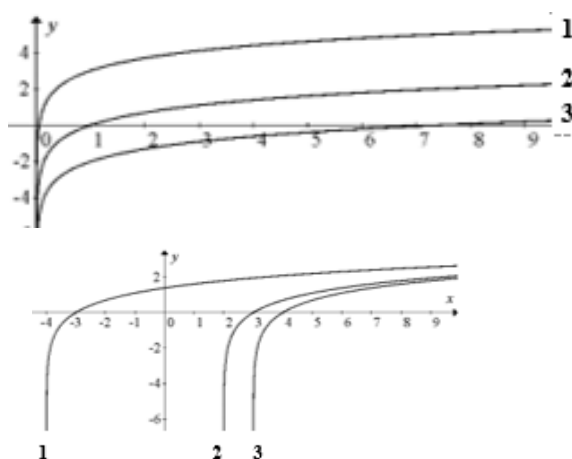
3. Sketch the graph of the following curves indicating:

- | | | |
|----------------------------|---|-----------------------|
| (i) x and y intercepts | (ii) equation of the horizontal asymptote | |
| (iii) domain and range | | |
| (a) $y = e^x$ | (b) $y = e^{2x}$ | (c) $y = 2e^x$ |
| (d) $y = e^x - 2$ | (e) $y = e^x + 1$ | (f) $y = e^x - 1$ |
| (g) $y = e^{x+2}$ | (h) $y = e^{x-1}$ | (i) $y = e^{x-3}$ |
| (j) $y = e^{x-2} + 1$ | (k) $y = e^{x+1} - 2$ | (l) $y = e^{x-4} - 3$ |

Key Idea - Graphs of Logarithmic Functions

1. For each of the following match the equation of the function with its graph.

- (a) (i) $y = \log_e x$
(ii) $y = \log_e(x) - 2$
(iii) $y = \log_e(x) + 3$
- (b) (i) $y = \log_e(x - 3)$
(ii) $y = \log_e(x - 2)$
(iii) $y = \log_e(x + 4)$



2. (a) Compared to the graph of $y = \log_e(x)$, the graph of $h(x)$ is shifted up 1 unit. Write the equation of $h(x)$.
(b) Compared to the graph of $y = \log_e(x)$, the graph of $h(x)$ is shifted to the left by 1 unit. Write the equation of $h(x)$.
(c) Compared to the graph of $y = \log_e(x)$, the graph of $h(x)$ is shifted 2 units downwards and 3 units to the right. Write the equation of $h(x)$.

3. Sketch the graph of the following curves indicating:

- | | | |
|-----------------------------|---|-----------------------------|
| (i) x and y intercepts | (ii) equation of the vertical asymptote | |
| (iii) domain and range | | |
| (a) $y = \log_e x$ | (b) $y = \log_e 2x$ | (c) $y = 2 \log_e x$ |
| (d) $y = \log_e x + 1$ | (e) $y = \log_e x - 2$ | (f) $y = \log_e x - 3$ |
| (g) $y = \log_e(x + 1)$ | (h) $y = \log_e(x - 2)$ | (i) $y = \log_e(x + 4)$ |
| (j) $y = \log_e(x - 5) + 2$ | (k) $y = \log_e(x + 2) - 4$ | (l) $y = \log_e(x + 1) + 2$ |